

User query:
"which object is the
closest to the microwave?"



recent video frames

frozen 3D
reconstruction



renderable from 97
discrete camera poses
(validity-checked)



explicit scene memory

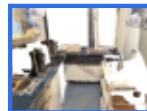
① Scene memory (once per scene)

direct answer from
frames + learned gate:
enough information?

Yes (71%):
return direct answer

Explore: start search

② Exploration gate



expand 3 actions per path,
keep 2 by calibrated confidence

walk deeper
(depth ≤ 3)



retained paths agree $\rightarrow$
stop early

cup $\checkmark$

③ Confidence-guided camera walks

final arbitration:
highest calibrated
confidence among
direct + explored
answers

(falls back to the
direct answer when
renders mislead)